
GENERAL RELATIVITY

Problem set 1, 11.09.2020 - solutions

■ PROBLEM 1 Problem 1.2 from Ref. [1]

Imagine that space (not spacetime) is actually a finite box, or in more sophisticated terms, a three-torus, of size L . By this, we mean that there is a coordinate system $x^\mu = (t, x, y, z)$ such that every point with coordinates (t, x, y, z) is *identified* with every point with coordinates $(t, x + L, y, z)$, $(t, x, y + L, z)$ and $(t, x, y, z + L)$. Note that the time coordinate is the same. Now consider two observers; observer A is at rest in this coordinate system (constant spatial coordinates), while observer B moves in the x -direction with constant velocity v . A and B begin at the same event, and while A remains still, B moves once around the universe and comes back to intersect the worldline of A without ever having to accelerate (since the universe is periodic). What are the relative proper times experienced in this interval by A and B ? Is this consistent with your understanding of Lorentz invariance?

SOLUTION: The proper times are related as, $T_B = T_A/\gamma$, $\gamma = 1/\sqrt{1 - (v/c)^2}$. Since Lorentz symmetry is broken by the periodicity, there is no inconsistency. There is an absolute rest frame, which can be established by maximizing L perceived by different inertial observers. To see how things work in detail, plot the space-time diagram and worldlines for both the observer at rest and the moving observer, with periodic boundary conditions properly incorporated. Note that this spacetime diagram looks very similar to the one for the twin paradox (with one subtle difference). What is that difference?

■ PROBLEM 2 Problem 1.4 from Ref. [1]

Projection effects can trick you into thinking that an astrophysical object is moving "superluminally". Consider a quasar that ejects gas with speed v at an angle θ with respect to the line-of-sight with angular speed $\frac{v_{app}}{D}$, where D is the distance to the quasar and v_{app} is the apparent speed. Derive an expression for v_{app} in terms of v and θ . Show that, for appropriate values of v and θ , v_{app} can be greater than 1.

SOLUTION: The gas' velocity can be decomposed into the tangential, $v_\perp = v \sin(\theta)$ and the component along the line-of-sight, $v_\parallel = v \cos(\theta)$. The angular velocity equals, $\omega = \Delta\alpha/\Delta T$, where $\Delta\alpha$ is the angle perceived by the observed in the interval ΔT as measured by the observer. From the simple triangular geometry we have,

$$\omega = \frac{v \sin(\theta) \Delta t}{D \Delta T}, \quad (2.1)$$

where Δt is the interval measured at the quasar. For $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, the gas will move towards the observer such that the time interval perceived by the observer will be shorter than that in the quasar frame (because the time that light takes to arrive shortens) by,

$$\Delta T = \left(1 - \frac{v}{c} \cos(\theta)\right) \Delta t \quad (2.2)$$

Inserting (2.2) into (2.1) gives,

$$\omega = \frac{v \sin(\theta)}{D \left(1 - \frac{v}{c} \cos(\theta)\right)} \equiv \frac{v_{\text{app}}}{D} \quad (2.3)$$

from which one obtains,

$$v_{\text{app}} = \frac{v \sin(\theta)}{1 - \frac{v}{c} \cos(\theta)}. \quad (2.4)$$

This can be greater than c if the right hand side is greater than c , or equivalently if,

$$\frac{v}{c} (\sin(\theta) + \cos(\theta)) = \sqrt{2} \frac{v}{c} \sin\left(\theta + \frac{\pi}{4}\right) > 1. \quad (2.5)$$

The sine function maximizes at $\theta = \frac{\pi}{4} \bmod[2\pi]$, at which the condition (2.5) is the weakest, $v > c/\sqrt{2}$. Conversely, in the limit when $v \rightarrow c$, the condition (2.5) is met for angles satisfying $\sin\left(\theta + \frac{\pi}{4}\right) > \frac{1}{\sqrt{2}}$, or equivalently, $\sin(2\theta) > 0$, from which we conclude $(0 < \theta < \frac{\pi}{2}) \bmod[\pi]$.

■ PROBLEM 3 Problem 1.5 from Ref. [1]

Particle physicists are so used to setting $c = 1$ that they measure mass in units of energy. In particular, they tend to use electron volts ($1 \text{ eV} = 1.6 \times 10^{-12} \text{ erg} = 1.8 \times 10^{-33} \text{ g}$), or, more commonly, keV, MeV, and GeV (10^3 eV , 10^6 eV and 10^9 eV , respectively). The muon has been measured to have a mass of 0.106 GeV and a rest frame lifetime of 2.19×10^{-6} seconds. Imagine that such a muon is moving in the circular storage ring of a particle accelerator, 1 kilometer in diameter, such that the muon's total energy is 1000 GeV . How long would it appear to live from the experimenter's point of view? How many radians would it travel around the ring?

SOLUTION: The energy of the muon is given by the Einstein's relation ($c = 1$), $E = \sqrt{m^2 + p^2} = m\gamma$, which follows from $\vec{p} = m\vec{u} = m\gamma\vec{v}$. We thus have, $\gamma = E/m \simeq 10000$, such that the lab frame life time of the muon is, $\tau_O = \gamma\tau = 0.0219 \text{ s}$. The circumference of the ring is $\pi \text{ km}$, which is equivalent to 2π radians. The muon is moving approximately with the speed of light $c \simeq 3 \times 10^5 \text{ km}$, thus in one decay time τ_O the muon traverses $\tau_O c \simeq 6600 \text{ km}$, which is about 13200 radians, or about 2100 times around the ring.

■ PROBLEM 4 Lorentz covariant formulation of electromagnetism.

The Maxwell action for electromagnetism is

$$S = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{4\pi}{c} j^\mu A_\mu \right], \quad (4.1)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor, A_μ is the vector potential, and $j^\mu = (c\rho, \vec{j})$ is the electromagnetic (charge) current. Furthermore,

$$F^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} F_{\alpha\beta} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & -B^3 & B^2 \\ E^2 & B^3 & 0 & -B^1 \\ E^3 & -B^2 & B^1 & 0 \end{pmatrix}, \quad (4.2)$$

where $\eta^{\alpha\beta} = \eta_{\alpha\beta} = \text{diag}(-1, 1, 1, 1)$ is the Minkowski metric tensor, and $\vec{E} = (E^1, E^2, E^3)$ and $\vec{B} = (B^1, B^2, B^3)$ are the electric and magnetic fields, respectively.

- (a) By making use of the variational principle, $\delta S/\delta A_\mu(x) = 0$, show that the implied equation of motion is

$$\partial_\nu F^{\nu\mu} = \frac{4\pi}{c} j^\mu . \quad (4.3)$$

Show further that (4.3) implies the usual inhomogeneous Maxwell's equations,

$$\nabla \cdot \vec{E} = 4\pi\rho , \quad -\frac{1}{c} \frac{\partial}{\partial t} \vec{E} + \nabla \times \vec{B} = \frac{4\pi}{c} \vec{j} . \quad (4.4)$$

- (b) Show that the dual field tensor,

$${}^*F^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} F_{\alpha\beta} = \begin{pmatrix} 0 & -B^1 & -B^2 & -B^3 \\ B^1 & 0 & E^3 & -E^2 \\ B^2 & -E^3 & 0 & E^1 \\ B^3 & E^2 & -E^1 & 0 \end{pmatrix} , \quad (4.5)$$

obeys a homogeneous equation,

$$\partial_\nu {}^*F^{\nu\mu} = 0 . \quad (4.6)$$

Here $\epsilon^{\mu\nu\alpha\beta}$ is the so-called Levi-Civita tensor, which is defined to be

$$\epsilon^{\mu\nu\alpha\beta} = \begin{cases} 1 & \text{if } (\mu\nu\alpha\beta) = \text{any even permutation of } (0123) , \\ -1 & \text{if } (\mu\nu\alpha\beta) = \text{any odd permutation of } (0123) , \\ 0 & \text{otherwise.} \end{cases} \quad (4.7)$$

Show further that these equations imply the homogeneous Maxwell's equations,

$$\nabla \cdot \vec{B} = 0 , \quad \frac{1}{c} \frac{\partial}{\partial t} \vec{B} + \nabla \times \vec{E} = 0 . \quad (4.8)$$

- (c) Recall that under Lorentz transformation $F^{\mu\nu}$ transforms as

$$F^{\mu\nu} \rightarrow F'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta F^{\alpha\beta} . \quad (4.9)$$

Consider first an ordinary rotation around the z -axis. Show that the magnetic and electric fields transform as ordinary vectors under such a rotation. Consider next a boost along the x -axis with velocity v . Show that we have the following transformation, mixing electric and magnetic fields ($\gamma = 1/\sqrt{1-\beta^2}$, $\beta = v/c$),

$$\begin{aligned} B'_1 &= B_1 , & B'_2 &= \gamma(B_2 + \beta E_3) , & B'_3 &= \gamma(B_3 - \beta E_2) , \\ E'_1 &= E_1 , & E'_2 &= \gamma(E_2 - \beta B_3) , & E'_3 &= \gamma(E_3 + \beta B_2) . \end{aligned} \quad (4.10)$$

Recall that a Lorentz boost along the x -axis is given by the transformation

$$x' = \gamma(x - \beta x^0) , \quad y' = y , \quad z' = z , \quad x'^0 = \gamma(x^0 - \beta x) . \quad (4.11)$$

- (d) Consider now an inertial system $\mathbf{I}$ and the Lorentz boosted system $\mathbf{I}'$ (where $\mathbf{I}'$ is not rotated relative to $\mathbf{I}$), moving with velocity $\vec{v}$ relative to $\mathbf{I}$. The formula generalizing formula (4.10) above is

$$\vec{B} = \gamma \vec{B}' + \frac{\vec{v}}{v^2} (\vec{v} \cdot \vec{B}') (1 - \gamma) + \gamma \frac{\vec{v}}{c} \times \vec{E}' , \quad (4.12)$$

$$\vec{E} = \gamma \vec{E}' + \frac{\vec{v}}{v^2} (\vec{v} \cdot \vec{E}') (1 - \gamma) - \gamma \frac{\vec{v}}{c} \times \vec{B}' . \quad (4.13)$$

Assume that $\vec{E}'$ and $\vec{B}'$ are constant and different from zero. Find the condition that $\vec{E}'$ and $\vec{B}'$ have to satisfy in order that there exists a $\vec{v}$ such that $\vec{E} = 0$, and find the corresponding $\vec{v}$ expressed in terms of $\vec{E}'$ and $\vec{B}'$. (Remark: by considering charged particles in this special Lorentz frame one easily finds how they will move.)

- (e) Show that the two quantities $\vec{E} \cdot \vec{B}$ and $|\vec{E}|^2 - |\vec{B}|^2$ are lorentz invariant, *i.e.* they are Lorentz scalars.

Hint: You might find the following identities useful,

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B}) , \quad (4.14)$$

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C} , \quad (4.15)$$

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{B} \cdot \vec{C})(\vec{A} \cdot \vec{D}) . \quad (4.16)$$

Try to prove these identities by using the antisymmetric symbol, ϵ^{ijk} . Recall that $(\vec{A} \times \vec{B})^i = \epsilon^{ijk} A^j B^k$.

SOLUTION: (a) Varying the action (4.1) with respect to A_μ gives,

$$\delta S = \int d^4x \left[-\frac{1}{2} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) F^{\mu\nu} - \frac{4\pi}{c} j^\mu \delta A_\mu \right] , \quad (4.17)$$

Renaming the dummy indices $\mu \leftrightarrow \nu$ in the first term yields a term that is equal to the second term. Integrating by parts the second term, and dropping a boundary term yields,

$$\delta S = \int d^4x \left[-\partial_\nu F^{\mu\nu} - \frac{4\pi}{c} j^\mu \right] \delta A_\mu(x) , \quad (4.18)$$

This ought to vanish for arbitrary δA_μ , such that the expression in square brackets must vanish, which yields the equation of motion (4.3).

(b) From Eqs. (4.5–4.6) we have,

$$\frac{1}{2} \epsilon^{\mu\nu\alpha\beta} \partial_\mu \partial_\alpha A_\beta - \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} \partial_\mu \partial_\beta A_\alpha = 0 . \quad (4.19)$$

The equality follows after one realizes that the derivatives in the first (second) term are symmetric in $\mu\alpha$ ($\mu\beta$), but the ϵ tensor is antisymmetric in the same pair of indices.

An inspection immediately gives that the 0th(spatial) components of Eq. (4.6) yields the first (second) homogeneous equation of motion in (4.8).

(c) The Lorentz matrix corresponding to the rotation around the z -axis is,

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) & 0 \\ 0 & \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} , \quad (4.20)$$

When this acts (twice) on $F^{\mu\nu}$ in Eq. (4.2) one obtains (in matrix notation),

$$F' = \Lambda \cdot F \cdot \Lambda^T = \begin{pmatrix} 0 & -(c_\theta E^1 - s_\theta E^2) & -(c_\theta E^2 + s_\theta E^1) & -E^3 \\ (c_\theta E^1 - s_\theta E^2) & 0 & -B^3 & (c_\theta B^2 + s_\theta B^1) \\ (c_\theta E^2 + s_\theta E^1) & B^3 & 0 & -(c_\theta B^1 - s_\theta B^2) \\ E^3 & -(c_\theta B^2 + s_\theta B^1) & (c_\theta B^1 - s_\theta B^2) & 0 \end{pmatrix} , \quad (4.21)$$

where we introduced a short-hand notation $c_\theta = \cos(\theta)$, $s_\theta = \sin(\theta)$. From (4.21) we see that (E^1, E^2) and (B^2, B^1) transform according to rotations while E^3 and B^3 do not transform, which is precisely what one would expect from rotations of vector fields.

The Lorentz matrix for the boosts in the x direction reads,

$$\tilde{\Lambda}(r) = \begin{pmatrix} \cosh(r) & -\sinh(r) & 0 & 0 \\ -\sinh(r) & \cosh(r) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \tilde{\Lambda}^{-1} = \begin{pmatrix} \cosh(r) & \sinh(r) & 0 & 0 \\ \sinh(r) & \cosh(r) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (4.22)$$

such that the transformed field tensor is (note that $\tilde{\Lambda}^T = \tilde{\Lambda}$ is symmetric),

$$F' = \tilde{\Lambda} \cdot F \cdot \tilde{\Lambda} = \begin{pmatrix} 0 & -E^1 & -\gamma(E^2 - \beta B^3) & -\gamma(E^3 + \beta B^2) \\ E^1 & 0 & -\gamma(B^3 - \beta E^2) & \gamma(B^2 + \beta E^3) \\ \gamma(E^2 - \beta B^3) & \gamma(B^3 - \beta E^2) & 0 & -B^1 \\ \gamma(E^3 + \beta B^2) & -\gamma(B^2 + \beta E^3) & B^1 & 0 \end{pmatrix}, \quad (4.23)$$

where we made use of, $\tanh(r) = v/c \equiv \beta$, $\cosh(r) = \gamma$, $\sinh(r) = \gamma v/c$. Comparing (4.23) with (4.10) one finds a perfect agreement.

(d) It is convenient to decompose the fields into their longitudinal and transverse components as follows,

$$\vec{E} = \vec{E}_{\parallel} + \vec{E}_{\perp}, \quad \vec{B} = \vec{B}_{\parallel} + \vec{B}_{\perp}, \quad (4.24)$$

where $\vec{E}_{\parallel} = \vec{v}(\vec{v} \cdot \vec{E})/\|\vec{v}\|^2$, $\vec{B}_{\parallel} = \vec{v}(\vec{v} \cdot \vec{B})/\|\vec{v}\|^2$, and analogously for the transformed fields. Eq. (4.12) can then be written as,

$$0 = \vec{E}'_{\parallel} + \gamma \left(\vec{E}'_{\perp} - \frac{\vec{v}}{c} \times \vec{B}'_{\perp} \right), \quad (4.25)$$

from which we see that,

$$\vec{E}'_{\parallel} = 0 \quad \text{and} \quad \vec{E}'_{\perp} = \frac{\vec{v}}{c} \times \vec{B}'_{\perp}. \quad (4.26)$$

This yields the following condition on the magnitude v of $\vec{v}$:

$$\frac{v}{c} = \frac{\|\vec{E}'_{\perp}\|}{\|\vec{B}'_{\perp}\|} < 1. \quad (4.27)$$

Since $\vec{E}'_{\parallel} = 0$ and $\vec{B}'_{\parallel}$ is unconstrained, this condition can be also interpreted as,

$$\|\vec{E}'\| < \|\vec{B}'\|. \quad (4.28)$$

It is not hard to convince oneself that conversely, a boosted frame in which $\vec{B} = 0$ exists if $\|\vec{B}'\| < \|\vec{E}'\|$.

(e) From Eqs. (4.12–4.13) we see that, out of the nine possible terms four vanish in the product $\vec{E} \cdot \vec{B}$, such that we have,

$$\begin{aligned} \vec{E} \cdot \vec{B} &= \gamma^2 \vec{E}' \cdot \vec{B}' + \frac{2(1-\gamma)}{v^2} (\vec{v} \cdot \vec{E}') (\vec{v} \cdot \vec{B}') + \frac{(1-\gamma)^2}{v^2} (\vec{v} \cdot \vec{E}') (\vec{v} \cdot \vec{B}') \\ &\quad - \frac{\gamma^2}{c^2} (\vec{v} \times \vec{E}') \cdot (\vec{v} \times \vec{B}'). \end{aligned} \quad (4.29)$$

Now, upon making use of $(c/v)^2 = \gamma^2/(\gamma^2 - 1)$ in the second and third terms and of Eq. (4.16) in the last term in (4.29), and combining all the terms one obtains, $\vec{E} \cdot \vec{B} = \vec{E}' \cdot \vec{B}'$.

To prove the invariance of $\vec{E}^2 - \vec{B}^2$, let us first consider the transformations of $\vec{E}^2$ and $\vec{B}^2$,

$$\vec{E}^2 = \gamma^2 \vec{E}'^2 + \left[\frac{2\gamma(1-\gamma) + (1-\gamma)^2}{v^2} \right] (\vec{v} \cdot \vec{E}')^2 - 2\gamma^2 \vec{E}' \cdot \left(\frac{\vec{v}}{c} \times \vec{B}' \right) + \gamma^2 \left(\frac{\vec{v}}{c} \times \vec{B}' \right)^2 \quad (4.30)$$

$$\vec{B}^2 = \gamma^2 \vec{B}'^2 + \left[\frac{2\gamma(1-\gamma) + (1-\gamma)^2}{v^2} \right] (\vec{v} \cdot \vec{B}')^2 + 2\gamma^2 \vec{B}' \cdot \left(\frac{\vec{v}}{c} \times \vec{E}' \right) + \gamma^2 \left(\frac{\vec{v}}{c} \times \vec{E}' \right)^2 \quad (4.31)$$

Notice first that the expressions in square brackets equal γ^2/c^2 . Next, one can use identities (4.14) and (4.16) in the third and fourth term in Eqs. (4.30) and (4.31), respectively, to finally obtain $\vec{E}^2 - \vec{B}^2 = \vec{E}'^2 - \vec{B}'^2$.

Since there is no obligation to prove Eqs. (4.14–4.16), we shall not provide proofs here.

References

- [1] Sean M. Carroll, “An introduction to General Relativity, Spacetime and Geometry” Addison Wesley (2004), ISBN 0-8053-8732-3.